

Lab 3: Component Modelling & Architectural Pattern Selection — Justification

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Project: Inter-City Freight Load Matching Marketplace

Architecture Selection

We chose **Microservices Architecture** for the Inter-City Freight Load Matching Marketplace System.

Reason 1: Independent Scaling

Load search and discovery traffic (carriers continuously browsing and filtering available loads) is far higher in volume than booking traffic, and spikes independently during peak posting hours. A microservices architecture lets the Load Posting & Search Service scale horizontally on its own, without over-provisioning the Bid & Booking Manager or Payment Service, which see comparatively steady, lower-volume transactional traffic.

Reason 2: Fault Isolation

The marketplace has functionally distinct workflows — posting, search, bidding, payment, and tracking — that don't need to fail together. If the Payment Service experiences an outage or the external payment gateway is slow, Shippers can still post loads and Carriers can still search and submit bids; only the final booking step is affected. In a monolithic or tightly layered architecture, a fault in one module risks bringing down the entire application.

Security Advantage

Isolating the Payment Service as its own component means only that service holds credentials and access to the payment gateway/PCI-sensitive data. The Load Posting, Search, and Tracking services never touch payment data, so a vulnerability or breach in any of those components cannot expose financial information — the blast radius of a compromise is contained to the single service that legitimately needs that access.

Performance Benefit

The Load Posting & Search Service can maintain its own read-optimized data store (e.g., indexed for route/date/cargo-type queries) without competing for database resources with the Bid & Booking Manager's write-heavy transactional workload (bids, bookings, payments). Separating these data access patterns into independent services prevents high-volume search queries from slowing down booking and payment operations, and vice versa.